

GLM

combined abundance/
detection model:
 $y_i \sim \text{Poisson}(\lambda_i)$

potential detection confounds:
spatial

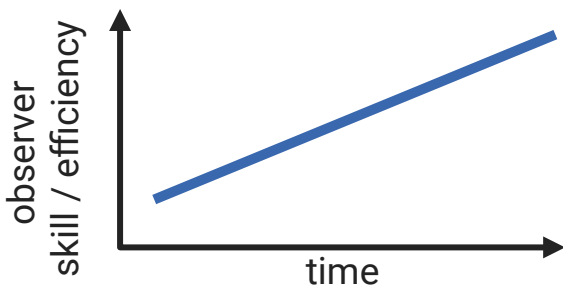


Habitat 1

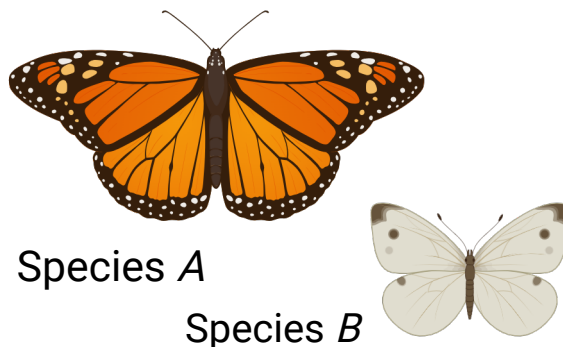


Habitat 2

temporal



taxonomic



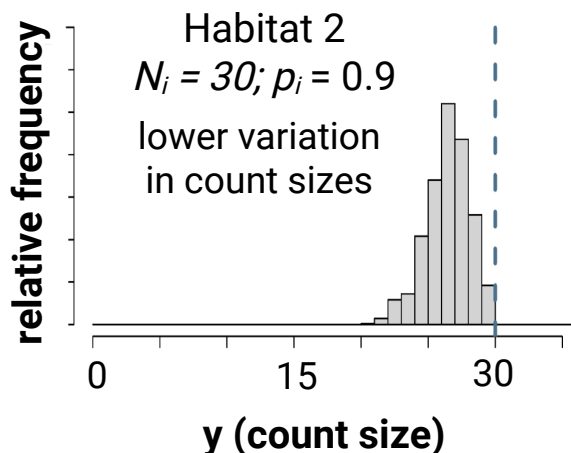
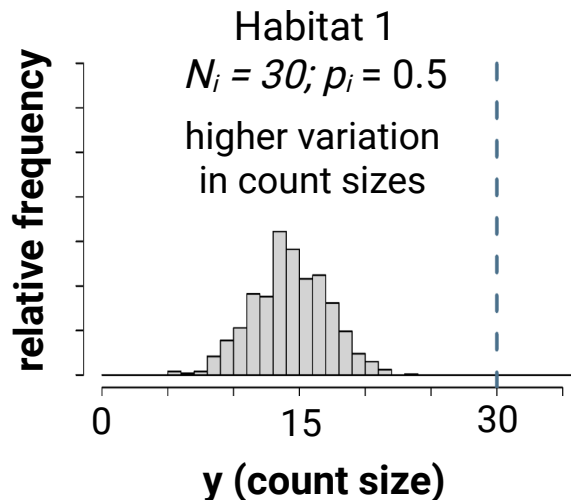
Binomial N-mixture

abundance model:

$$N_i \sim \text{Poisson}(\lambda_i)$$

detection model:

$$y_{i,j} \sim \text{binomial}(N_i, p_i)$$



Multinomial N-mixture

abundance model:

$$N_i \sim \text{Poisson}(\lambda_i)$$

detection model:

$$n_i \sim \text{binomial}(N_i, 1 - \pi_0);$$

$$y_i | n_i \sim \text{multinomial}(n_i, \pi_i^c)$$

